Introduction to Intelligent Systems (CAP0055)

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Q1. Design a fuzzy logic system to determine the washing machine cycle time based on :

a. Dirt level

b. Load size

Requirements :

a. Define fuzzy sets for Dirt Level (Low, Medium, High).

b. Define fuzzy sets for Load Size (Small, Medium, Large).

c. Create at least 6 fuzzy rules.

d. Implement the system using MATLAB Fuzzy Logic Toolbox.

e. Show the membership functions and rule viewer output.

Ans. Inputs :

- Dirt Level (Low, Medium, High)
- Load Size (Small, Medium, Large)

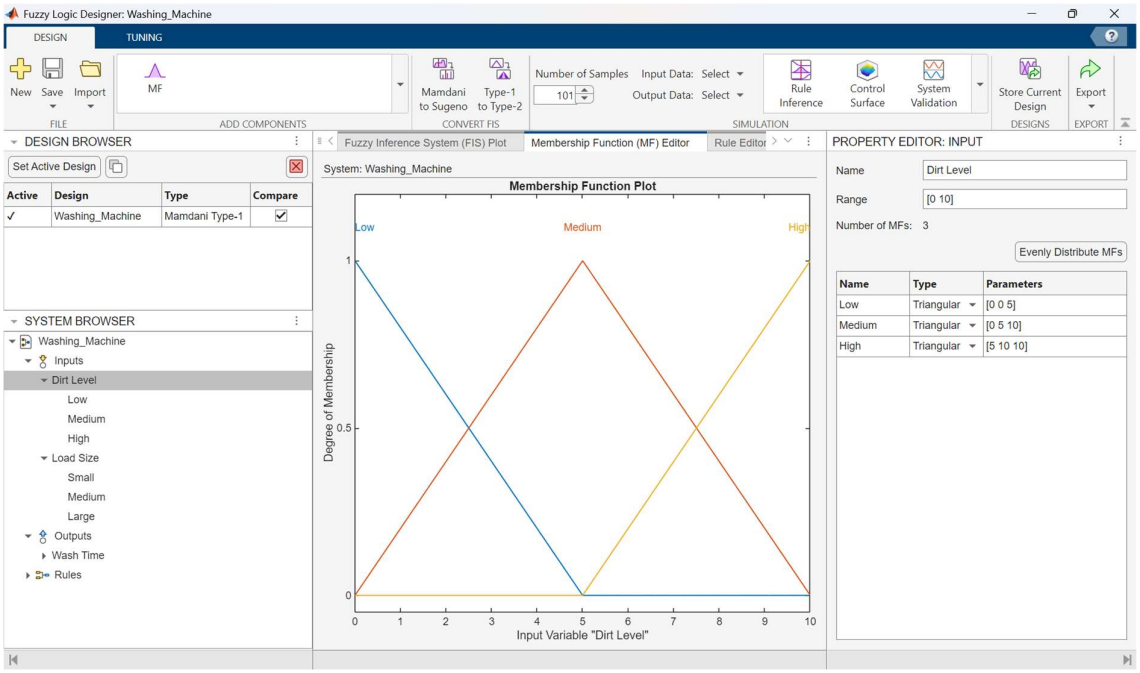
Output :

- Wash Time (Short, Medium, Long)

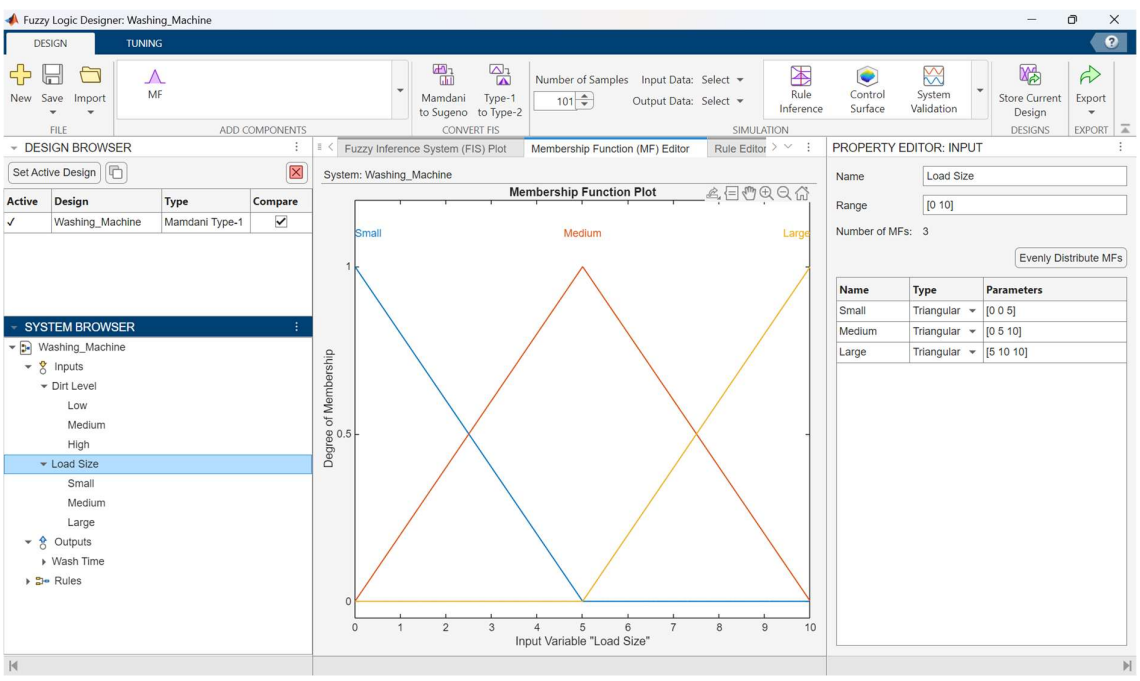
Membership Functions :

Triangular membership functions are used for all variables to represent linguistic values.

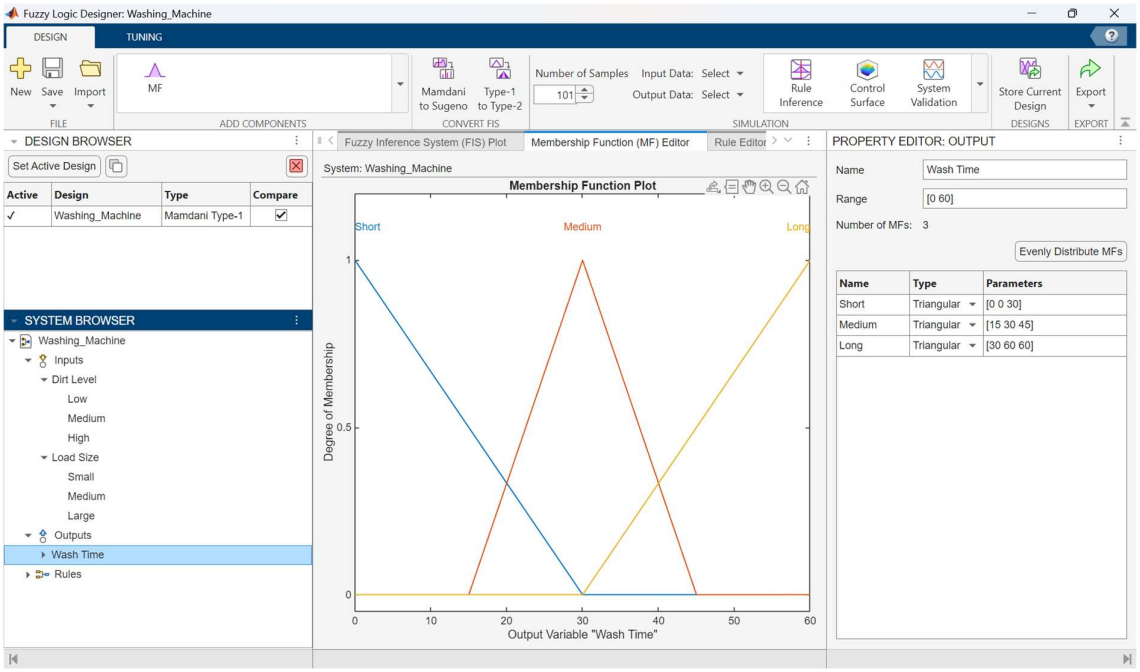
1) Dirt Level



2) Load Size



3) Wash Time



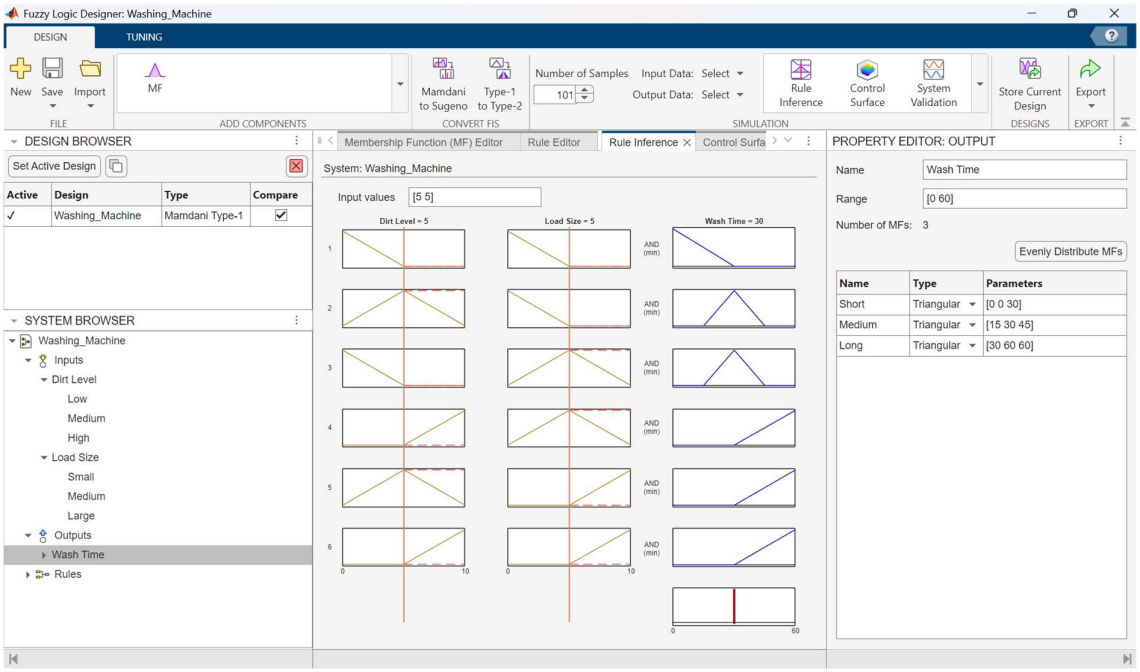
Fuzzy Rules :

1) IF Dirt is Low AND Load is Small then Time is Short
2) IF Dirt is Low AND Load is Medium then Time is Medium
3) IF Dirt is Medium AND Load is Small then Time is Medium
4) IF Dirt is Medium AND Load is Large then Time is Long
5) IF Dirt is High AND Load is Medium then Time is Long
6) IF Dirt is High AND Load is Large then Time is Long

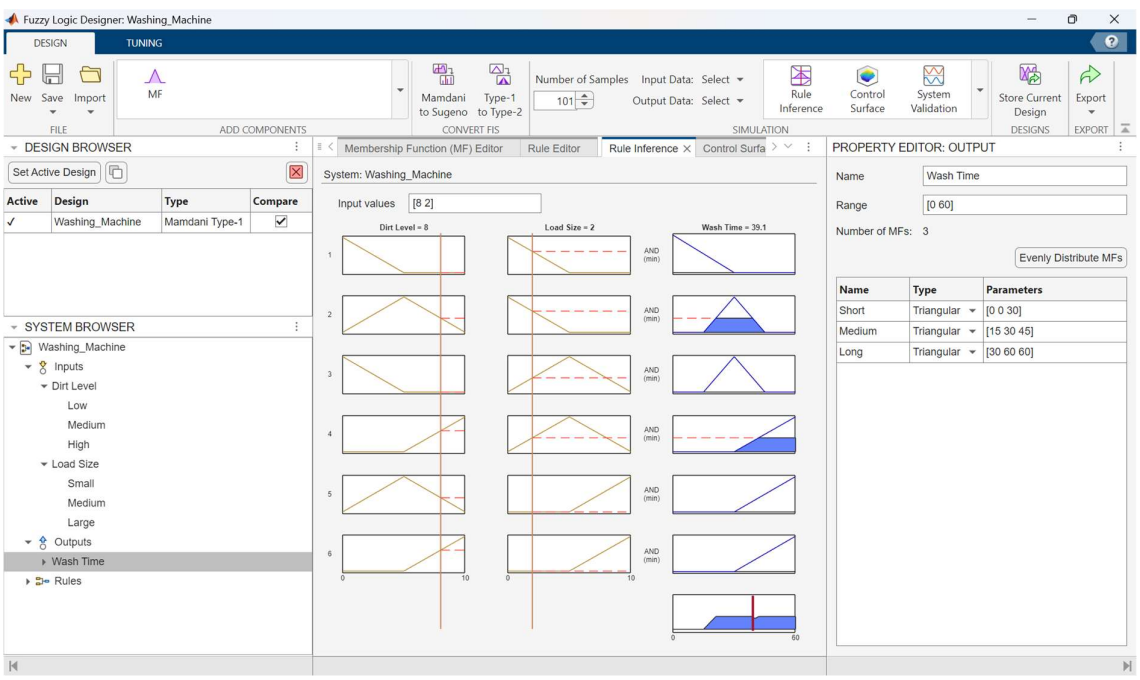
Rule Viewer :

The rule viewer shows how different input values affect the output based on fuzzy rules.

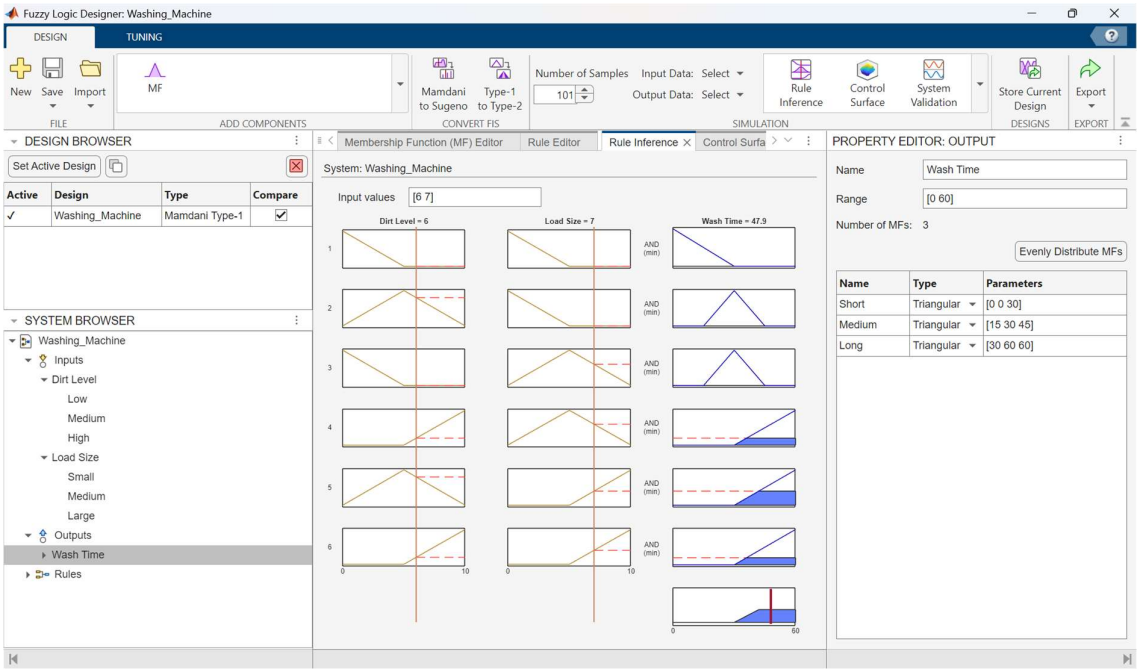
1) Input - [5 5], Output – 30



2) Input – [8 2], Output – 39.1

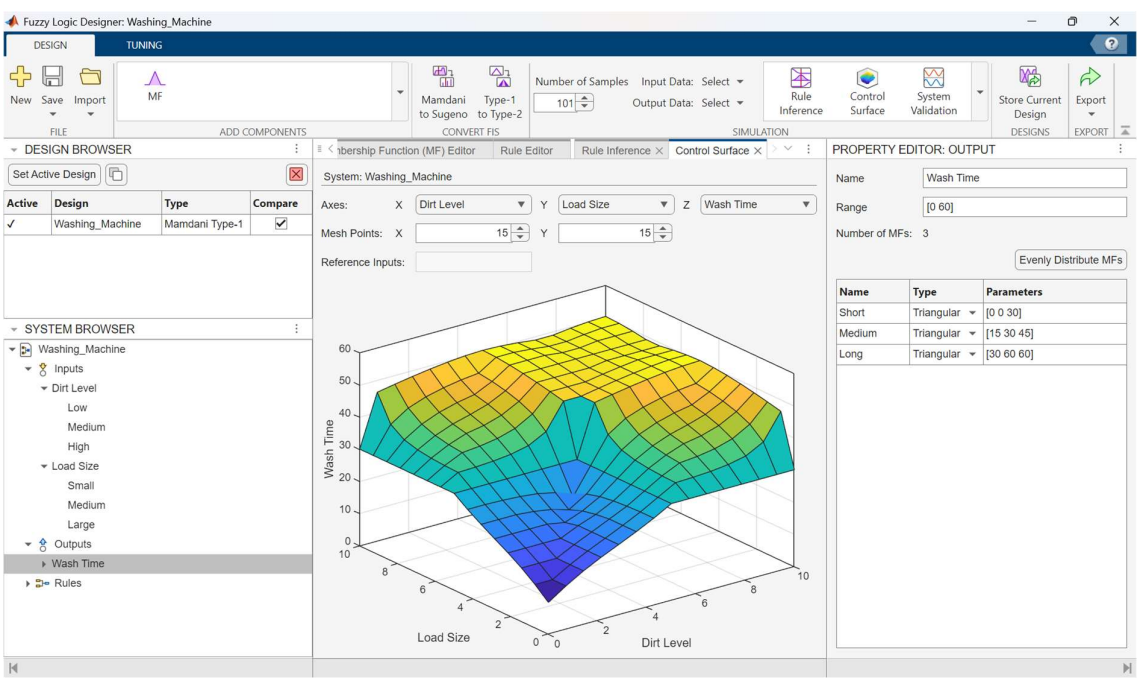


3) Input – [6 7], Output – 47.9



Surface Viewer :

The surface viewer shows the relationship between dirt level, load size, and wash time in a 3D graph.



Q2. Develop a hybrid intelligent system combining fuzzy logic and neural networks to predict student performance level.

Requirements :

a. Inputs: Attendance, Assignment Marks, Test Marks

b. Output: Performance Level (Poor, Average, Good)

c. Explain how fuzzy rules and neural network learning are integrated.

Ans. Inputs :

- Attendance
- Assignment Marks
- Test Marks

Output :

- Performance Level (Poor, Average, Good)

Methodology :

- The system uses ANFIS (Adaptive Neuro-Fuzzy Inference System).
- Initially, fuzzy logic defines rules and membership functions.
- Then, the neural network component trains the system using data.
- During training, the system adjusts membership functions and rule parameters to minimize error.
- Thus, fuzzy logic provides reasoning and neural networks provide learning.

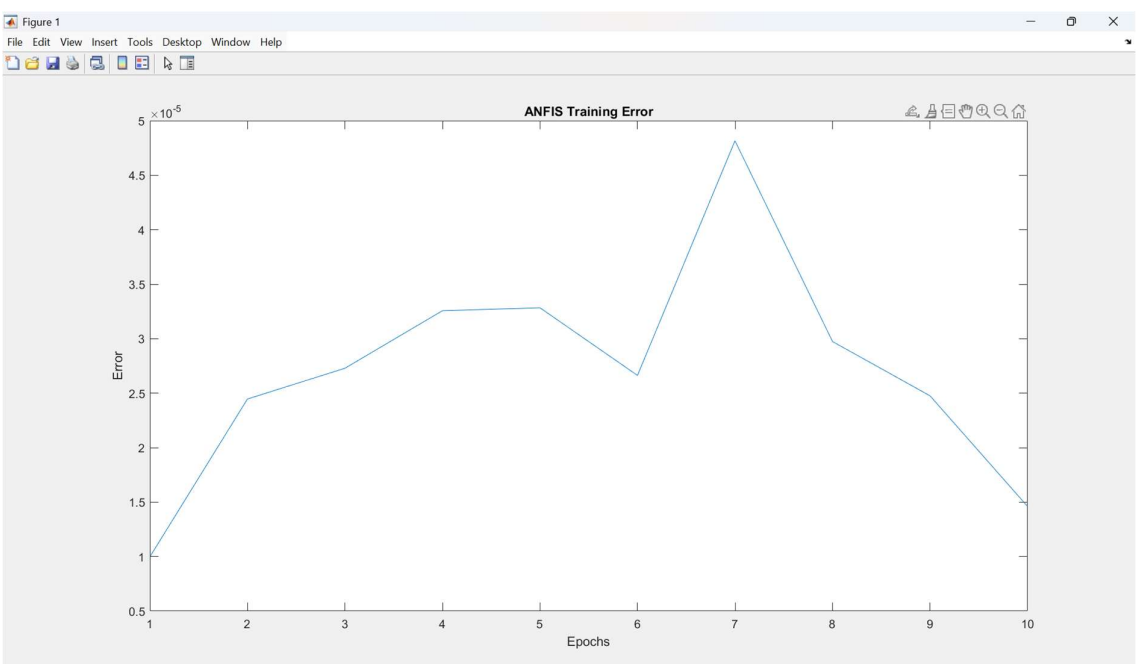
Dataset :

The dataset consists of student records including attendance, assignment marks, test marks, and performance scores.

Attendance	Assignment	Test	Performance
80	75	70	85
60	65	60	65
40	50	45	50
90	85	88	90
30	40	35	40
70	60	65	70
50	55	60	60
85	80	82	88

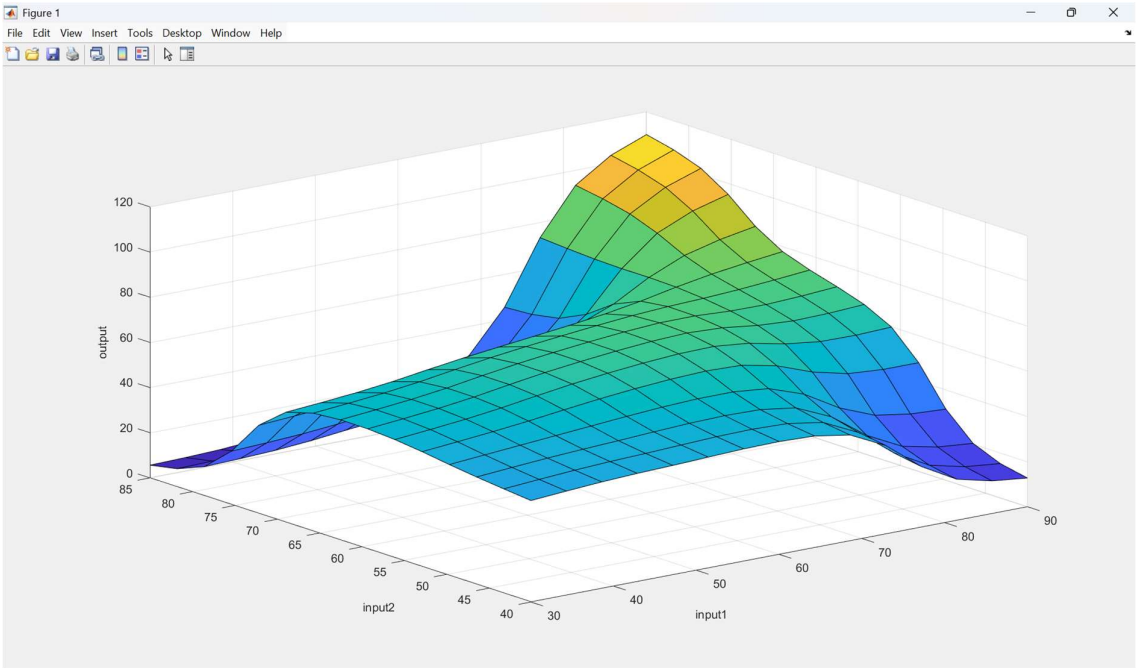
Training Output :

The ANFIS model is trained over multiple epochs. The error decreases with each iteration, showing learning.



Surface Viewer :

The surface plot shows the relationship between inputs and predicted performance.



Performance Levels :

Performance is categorized as :

0–50 is Poor

50–75 is Average

75–100 is Good